

Supplementary Information: Payload-Aware Autonomous ADC Linker Design

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1 Full context ranking (all 38 contexts)

Handle	Trigger	Payload	mean	best	SD	seeds
Oxyamine	Non-cleavable	cytotoxin	0.92	0.93	0.01	2
Tetrazine	Non-cleavable	cytotoxin	0.90	0.91	0.02	2
Bromoacetamide	Non-cleavable	cytotoxin	0.85	0.87	0.01	2
DBCO	Non-cleavable	immunomodulator	0.78	0.82	0.04	2
NHS_ester	Non-cleavable	cytotoxin	0.82	0.82	0.00	2
DBCO	Non-cleavable	cytotoxin	0.76	0.81	0.05	2
DBCO	Non-cleavable-rigid	immunomodulator	0.79	0.80	0.01	2
DBCO	Non-cleavable	oligonucleotide	0.79	0.80	0.01	2
DBCO	Non-cleavable-rigid	cytotoxin	0.78	0.78	0.01	2
DBCO	Non-cleavable-rigid	oligonucleotide	0.75	0.76	0.01	2
Maleimide	Non-cleavable	cytotoxin	0.72	0.75	0.02	2
Maleimide	Non-cleavable	immunomodulator	0.72	0.74	0.02	2
TCO	Non-cleavable	cytotoxin	0.68	0.70	0.02	2
Maleimide	Non-cleavable-rigid	oligonucleotide	0.68	0.68	0.00	2
Maleimide	Non-cleavable	oligonucleotide	0.67	0.68	0.01	2
Maleimide	Non-cleavable-rigid	immunomodulator	0.67	0.68	0.00	2
Maleimide	Non-cleavable-rigid	cytotoxin	0.67	0.67	0.00	2
Tetrazine	Val-Cit-PABC	cytotoxin	0.59	0.61	0.03	2
Bromoacetamide	Val-Cit-PABC	cytotoxin	0.58	0.60	0.02	2
Oxyamine	Val-Cit-PABC	cytotoxin	0.59	0.60	0.01	2
NHS_ester	Val-Cit-PABC	cytotoxin	0.58	0.59	0.01	2
Disulfide	Non-cleavable	cytotoxin	0.57	0.59	0.02	2
TCO	Val-Cit-PABC	cytotoxin	0.56	0.56	0.00	2
DBCO	Val-Cit-PABC	immunomodulator	0.53	0.55	0.02	2
DBCO	Val-Cit-PABC	cytotoxin	0.51	0.53	0.02	2
Maleimide	Val-Cit-PABC	immunomodulator	0.52	0.52	0.00	2
Maleimide	Val-Cit-PABC	cytotoxin	0.51	0.51	0.00	2
Disulfide	Val-Cit-PABC	cytotoxin	0.42	0.43	0.02	2
DBCO	Val-Cit-PABC	oligonucleotide	0.38	0.40	0.02	2
Maleimide	Val-Cit-PABC	oligonucleotide	0.35	0.36	0.01	2
Maleimide	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Bromoacetamide	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Oxyamine	Glucuronide	cytotoxin	0.08	0.08	0.00	2
NHS_ester	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Tetrazine	Glucuronide	cytotoxin	0.08	0.08	0.00	2
DBCO	Glucuronide	cytotoxin	0.08	0.08	0.00	2
TCO	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Disulfide	Glucuronide	cytotoxin	0.07	0.07	0.00	2

2 Payload-class re-scoring (mean composite by payload rule)

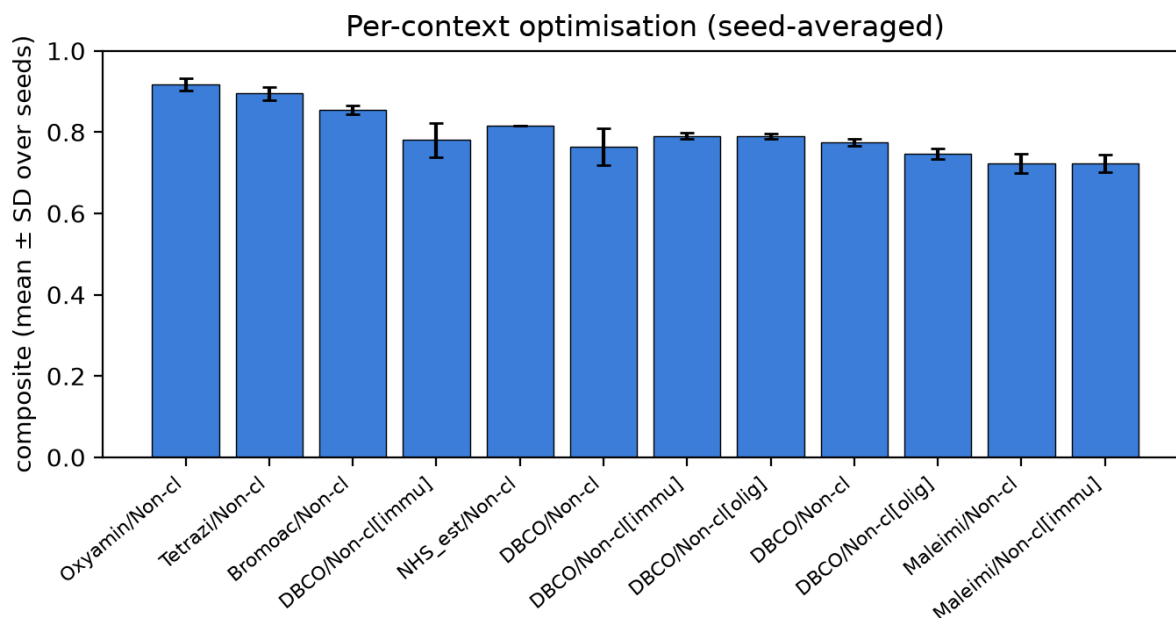


Figure 1: Per-context optimisation, mean $\pm$ SD over seeds (top contexts). Modest seed variance supports the ranking.

Handle	Trigger	cytotoxin	oligonucleotide	immunomodulator
DBCO	Non-cleavable	0.76	0.79	0.78
DBCO	Non-cleavable-rigid	0.78	0.75	0.79
Maleimide	Non-cleavable	0.72	0.67	0.72
Maleimide	Non-cleavable-rigid	0.67	0.68	0.67
DBCO	Val-Cit-PABC	0.51	0.38	0.53
Maleimide	Val-Cit-PABC	0.51	0.35	0.52

3 Table S1: Stability-surrogate calibration

Mechanistic class	stability	expected
acid-labile hydrazone	0.00	low
redox disulfide	0.80	intermediate
protease dipeptide	0.85	high
non-cleavable thioether	0.85	high

4 Commercial linker benchmark

Name	ADC / class	composite	QED	MW
mc-Val-Cit-PABC	brentuximab vedotin (Ad...	0.37	0.116	573
mc-Val-Ala-PABC	Val-Ala class	0.40	0.244	487
mc-GGFG	trastuzumab deruxtecan ...	0.39	0.128	530
MCC (non-cleavable)	trastuzumab emtansine (...)	0.59	0.737	237
SPDB disulfide	mirvetuximab soravtansi...	0.62	0.600	229
sulfo-SPDB	hindered disulfide class	0.66	0.444	309
AcBut hydrazone	gemtuzumab ozogamicin (...)	0.00	0.340	236
beta-glucuronide	glucuronide linker class	0.00	0.156	508
maleimidocaproyl (mc)	mc handle	0.59	0.514	211

5 Table S2: Top designed linkers (full SMILES)

Context	Payload	Score	SMILES
Tetraz/Non-cl	cytot	0.87	<chem>Cc1nnc(Nc2cc(S(=O)(=O)O)cc(S(=O)(=O)NCc3cccc3)c2)nn1</chem>
Tetraz/Non-cl	cytot	0.87	<chem>Cc1nnc(Nc2ccc(S(=O)(=O)O)c(S(=O)(=O)NCc3cccc3)c2)nn1</chem>
Tetraz/Non-cl	cytot	0.87	<chem>Cc1nnc(Nc2ccc(S(=O)(=O)NCc3cccc3)cc2N)nn1</chem>
Bromoa/Non-cl	cytot	0.87	<chem>O=C(CBr)NS(=O)(=O)NC(=O)CCS(=O)(=O)NCc1cccc1</chem>
Bromoa/Non-cl	cytot	0.86	<chem>O=C(CBr)NS(=O)(=O)c1cnccc1C(=O)NS(=O)(=O)NCc1cccc1</chem>
Bromoa/Non-cl	cytot	0.85	<chem>O=C(CBr)NS(=O)(=O)c1ccc(S(=O)(=O)NCc2cccc2)cc1S(=O)(=O)O</chem>
Bromoa/Non-cl	cytot	0.85	<chem>O=C(CBr)NS(=O)(=O)c1ccc(S(=O)(=O)O)c(S(=O)(=O)NCc2cccc2)c1</chem>
Bromoa/Non-cl	cytot	0.85	<chem>O=C(CBr)NS(=O)(=O)Nc1cnccc1S(=O)(=O)NCc1cccc1</chem>

6 Table S3: Boltz-2 co-folding with cathepsin B (full panel)

Linker	type	ipTM	pLDDT	Kd (nM)
Maleimide/Val-Cit-PABC	designed	0.85	—	45.0
Maleimide/Non-cleavable-r...	designed	0.25	—	999.0

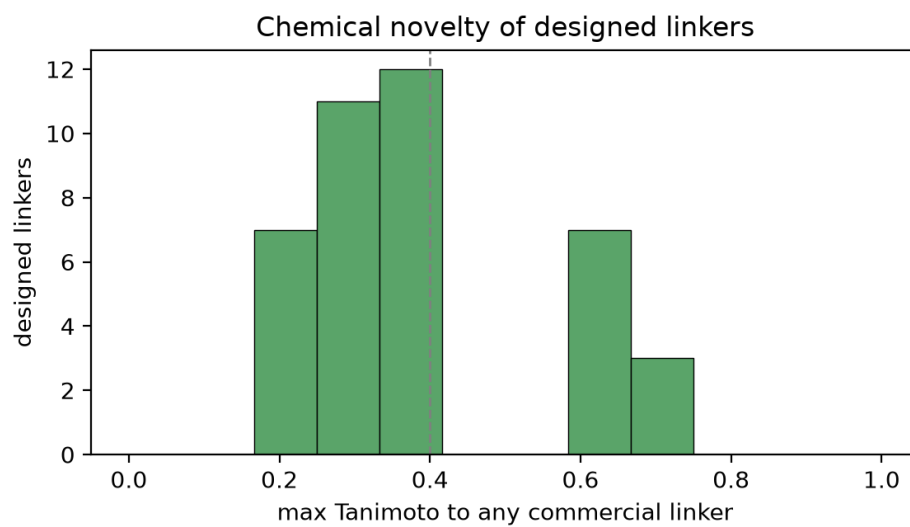


Figure 2: Chemical novelty: nearest-neighbour Tanimoto of each designed linker to any commercial linker. Most designs are distinct chemotypes (similarity < 0.4), not rediscoveries.